

Article

Reconstructing Quantum States from Sparse Measurements

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Abstract: Quantum state tomography (QST) is a central technique to fully characterize an unknown quantum state. However, standard QST requires an exponentially growing number of quantum measurements against the system size, which limits its application to smaller systems. Here, we explore the sparsity of underlying quantum state and propose a QST scheme that combines the matrix product states' representation of the quantum state with a supervised machine learning algorithm. Our method could reconstruct the unknown sparse quantum states with very high precision using only a portion of the measurement data in a randomly selected basis set. In particular, we demonstrate that the Wolfgang states could be faithfully reconstructed using around 25% of the whole basis, and that the randomly generated quantum states, which could be efficiently represented as matrix product states, could be faithfully reconstructed using a number of bases that scales sub-exponentially against the system size.

Keywords: quantum state tomography; matrix product state; sparse measurements

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1. Introduction

Characterizing unknown quantum states is of central importance in quantum-information processing. Standard quantum state tomography (QST) works by measuring the unknown quantum state in a set of informationally complete bases, the complexity of which grows exponentially with the system size [1,2]. As a result, the standard QST can only be feasibly implemented for a very small number of qubits.

Various methods have been proposed to scale up QST to larger systems, usually with additional assumptions regarding the underlying quantum state, for example, explicitly making use of the sparsity [3–6] or the symmetry [7,8] of the unknown quantum state, or assuming that the unknown quantum state is generated by a low-depth quantum circuit [9,10]. Importantly, for (fairly) pure quantum states with a bounded bipartition entanglement, there are polynomial reconstruction algorithms that go against the system size, with guaranteed success [11–14], as well as heuristic methods that could be scalable to large systems [15–19], based on the matrix product state (MPS) representation of the quantum state. Another class of QST methods that shows great potential of scalability in practice is neural-network-based QST methods [20–31], where the unknown quantum states are (indirectly) represented as neural networks whose parameters are variationally optimized based on the measurement outcomes. Additionally, the complexity of QST could also be greatly reduced if one only aims for partial information [32–35].

The existing scalable QST schemes based on MPSs or neural networks are mostly based on a set of measurement outcomes (bitstrings), which are probably the most friendly data for real experiments. In this work, instead of using measured bitstrings as the training data, we use the expectation values of a set of observables as our training data. Concretely, we

generate a set of random product observables and compute the expectation values of these observables for the unknown quantum state (ideally, this should be done experimentally, but here we will use synthetic data). The set of product observables, together with their expectation values, are then taken as our training data to reconstruct the target quantum state. This approach is similar to the standard QST, with the difference that we only use a portion of all the possible product observables instead of all the possible ones. To faithfully reconstruct the unknown quantum state without the problem of under fitting, we need to use a variational ansatz for the quantum state, whose number of tunable parameters does not grow exponentially with the system size, and here we will use MPS as our variational ansatz.

Compared to the QST schemes, which directly use the measured bitstrings as training data, our approach would likely require more training data to evaluate the expectation value of each product observable accurately; to do this, one needs to generate a larger number of bitstrings in the corresponding basis in general. However, a possible advantage of our approach is that it might be able to reconstruct the underlying quantum state with a precision close to the standard QST, and that it could be easily used in combination with certain error-mitigation schemes to make it less error-prone [36,37]. Additionally, in many experiments the expectation values of a certain class of observables are of the most interest, such as the local observables, and our approach is able to give the “best” quantum state that fits for those observables (but may not be exactly the same as the true quantum state).

The paper is organized as follows. In Section 2, we show the details of our algorithm, including the construction of the MPS ansatz as well as the variational optimization. In Section 3, we demonstrate our algorithm by applying it to reconstruct the Greenberger–Horne–Zeilinger (GHZ) state, Wolfgang (W) states, and randomly generated MPSs. We conclude in Section 4.

2. Method

The standard QST works by measuring identically prepared quantum states in an informationally complete basis. In case of a single qubit, the corresponding quantum state lives on the Bloch sphere and can be written as

$$\hat{\rho} = \frac{1}{2}(I_2 + p_x\sigma_x + p_y\sigma_y + p_z\sigma_z), \quad (1)$$

where I_2 is the 2×2 identity matrix, $\sigma_x, \sigma_y, \sigma_z$ are the Pauli matrices, and the coefficients $p_j = \text{tr}(\sigma_j\hat{\rho})$ ($j \in \{x, y, z\}$) are the expectation values of the Pauli matrices. By measuring the quantum state $\hat{\rho}$ and computing the expectation values p_j , we can recover the unknown quantum state $\hat{\rho}$. The matrices $\{I_2, \sigma_x, \sigma_y, \sigma_z\}$ form a complete basis for the single-qubit quantum state $\hat{\rho}$, and we will denote $\sigma_0 = I_2, \sigma_1 = \sigma_x, \sigma_2 = \sigma_y, \sigma_3 = \sigma_z$ for notational convenience in the next. Then, all the Pauli strings defined as

$$\hat{P}_{i_1, i_2, \dots, i_n} = \sigma_{i_1} \otimes \sigma_{i_2} \otimes \dots \otimes \sigma_{i_n}, \quad (2)$$

with $i_n \in \{0, 1, 2, 3\}$ form a complete basis for the case of a multi-qubit quantum system, namely, a general n -qubit quantum state can be written as

$$\hat{\rho} = \frac{1}{2^n} \sum_{i_1, i_2, \dots, i_n=0}^3 p_{i_1, i_2, \dots, i_n} \hat{P}_{i_1, i_2, \dots, i_n}, \quad (3)$$

The free parameters (all the coefficients $p_{i_1, i_2, \dots, i_n}$) required to determine general $\hat{\rho}$ scales as 4^n as can be seen from Equation (3) [1,2].

In this work, we propose a method to reconstruct the unknown quantum state using only a randomly selected portion of all the possible coefficients, the number of which is dependent on the underlying quantum state and does not necessarily grow exponentially as n increases. A similar idea has also been explored in compressive-sensing-based

QST [3–6], which reconstructs an unknown density matrix with most of the entries to be 0. The difference here is that we assume that the unknown quantum state can be efficiently parameterized by an MPS instead of a sparse matrix. We note that, in general, an MPS is a more sparse representation than a sparse matrix since the number of free parameters in an MPS only grows polynomially with n , while a sparse matrix could often grow exponentially as 2^n (but not 4^n).

Now, we describe our algorithm in detail. In this work, we will only focus on pure quantum states, but our method can be easily generalized to mixed quantum states as well. We assume that the unknown pure quantum state (the target state) is denoted as $|\phi\rangle$:

$$|\phi\rangle = \sum_{s_1 s_2 \dots s_n} \phi_{s_1 s_2 \dots s_n} |s_1 s_2 \dots s_n\rangle. \quad (4)$$

Our goal is to reconstruct a quantum state $|\tilde{\phi}\rangle$, whose coefficient tensor $\tilde{\phi}_{s_1 s_2 \dots s_n}$ can be written as an MPS

$$\tilde{\phi}_{s_1 s_2 \dots s_n} = \sum_{a_0 a_1 \dots a_n} M_{a_0 a_1}^{s_1} M_{a_1 a_2}^{s_2} \dots M_{a_{n-1} a_n}^{s_n}, \quad (5)$$

and that $|\tilde{\phi}\rangle$ could be a good approximation for $|\phi\rangle$. As shown in Equation (5), an MPS is the product of a chain of rank-3 tensors, where s_l denotes the “physical” index and a_l denotes the “auxiliary” index ($\dim(a_0) = \dim(a_n) = 0$ are trivial auxiliary indices added at the boundaries for notational convenience). The size of the MPS (the number of parameters) is characterized by the size of the auxiliary indices, denoted as

$$D = \max_{0 \leq l \leq n} \dim(a_l), \quad (6)$$

which is referred to as the *bond dimension*. In general, any quantum state can be transformed into an MPS using a series of singular value decompositions (SVDs) as long as D is large enough, as shown in Figure 1a, and a quantum state can be efficiently written as an MPS if D is almost a constant as n grows. With a fixed bond dimension D , we can see that the number of parameters in Equation (5) is approximately $2nD^2$, which only grows linearly as n increases. For general quantum states, the bond dimension D required could also grow exponentially, and one may need an exponentially growing number of training data (in the worst case, one may need the whole training data as in the standard QST) to ensure the accurate reconstruction of the unknown state.

Our algorithm variationally updates the parameters in the MPS ansatz in Equation (5), using a training dataset that is a randomly selected portion of all the possible coefficients in Equation (3). In the following, we use a slightly different but equivalent local basis set as follows:

$$\begin{aligned} m_0 &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, & m_1 &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \\ m_2 &= \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, & m_3 &= \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}. \end{aligned} \quad (7)$$

We note that the matrices m_1 and m_2 are not Hermitian; nevertheless, such observables could also be easily measured in experiments [38]. To generate the training data, we randomly select M different measurement operators; each of them is denoted as

$$\hat{O}_j = m_{i_1}^{j_1} \otimes m_{i_2}^{j_2} \otimes \dots \otimes m_{i_n}^{j_n}, \quad (8)$$

where $1 \leq j \leq M$ and $i_l \in \{0, 1, 2, 3\}$ for $1 \leq l \leq n$; then, we assume that the expectation values of each $\hat{O}_j$ has already been obtained, which is written as

$$y_j = \langle \phi | \hat{O}_j | \phi \rangle, \quad (9)$$

and we will take all the pairs $(\hat{O}_j, y_j)$ as our training data. Ideally, the training data $(\hat{O}_j, y_j)$ are obtained experimentally. In our numerical experiment, we will first represent the target quantum state $|\phi\rangle$ as an MPS and then generate synthetic training data by computing those expectation values in Equation (9) on $|\phi\rangle$ using standard MPS techniques [39].

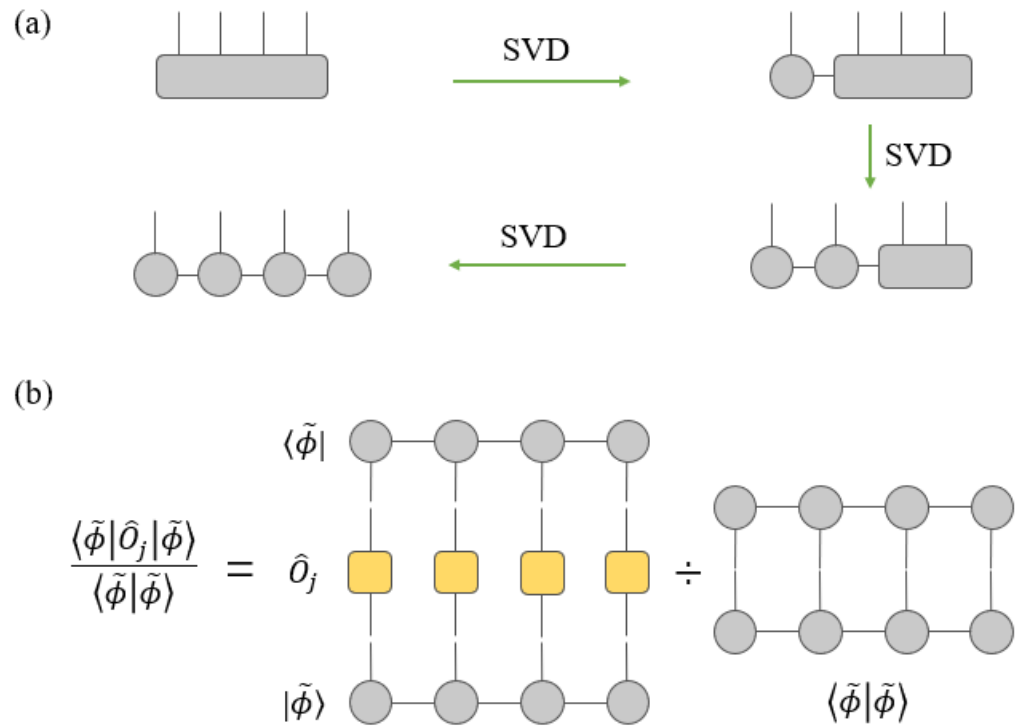


Figure 1. (a) Decomposition of a four-qubit quantum state into a matrix product state using a series of singular value decompositions. (b) Evaluating the expectation value of a Pauli string $\hat{O}_j$ on the variational MPS ansatz $|\tilde{\phi}\rangle$.

In the training stage, we will take Equation (5) as our variational ansatz and then try to optimize the parameters of it such that the expectation values of the randomly selected operators $\hat{O}_j$ are closest to their ground truth values. Concretely, we use the mean square loss as our loss function, denoted as

$$\text{Loss}(\tilde{\phi}) = \frac{1}{M} \sum_{j=1}^M \left(\frac{\langle \tilde{\phi} | \hat{O}_j | \tilde{\phi} \rangle}{\langle \tilde{\phi} | \tilde{\phi} \rangle} - y_j \right)^2, \quad (10)$$

where $\langle \tilde{\phi} | \hat{O}_j | \tilde{\phi} \rangle$ can be efficiently evaluated as

$$\begin{aligned} \langle \tilde{\phi} | \hat{O}_j | \tilde{\phi} \rangle &= \sum_{\substack{a_0, \dots, a_n \\ a'_0, \dots, a'_n}} \left(\sum_{s_1 s'_1} M_{a_0 a_1}^{s_1} (m_{i_1}^j)^{s_1 s'_1} M_{a'_0 a'_1}^{s'_1} \right) \\ &\quad \cdots \times \left(\sum_{s_n s'_n} M_{a_{n-1} a_n}^{s_n} (m_{i_n}^j)^{s_n s'_n} M_{a'_{n-1} a'_n}^{s'_n} \right) \end{aligned} \quad (11)$$

and $\langle \tilde{\phi} | \tilde{\phi} \rangle$ can be efficiently evaluated as

$$\begin{aligned} \langle \tilde{\phi} | \tilde{\phi} \rangle = & \sum_{\substack{a_0, \dots, a_n \\ a'_0, \dots, a'_n}} \left(\sum_{s_1 s'_1} M_{a_0 a_1}^{s_1} M_{a'_0 a'_1}^{s'_1} \right) \\ & \dots \times \left(\sum_{s_n s'_n} M_{a_{n-1} a_n}^{s_n} M_{a'_{n-1} a'_n}^{s'_n} \right). \end{aligned} \quad (12)$$

The evaluation of Equations (11) and (12) is also demonstrated in Figure 1b. In our loss function in Equation (10), we have divided by $\langle \tilde{\phi} | \tilde{\phi} \rangle$ since, in general, our ansatz quantum state $|\tilde{\phi}\rangle$ does not have to be normalized. Moreover, compared to most MPS-based algorithms where the underlying MPS is kept in a canonical form and the optimization is done for each site tensors of it one by one, we will keep the MPS in the general form and directly optimize all of the site tensors using a gradient-based optimization algorithm, and the gradient of the loss function in Equation (10) will be directly calculated using automatic differentiation as in Ref. [40]. Our algorithm is briefly summarized in Algorithm 1.

Algorithm 1 QST from sparse measurements.

- 1: Randomly selecting M measurement operators $\hat{O}_j$ and then performing quantum measurements on the unknown quantum state $|\phi\rangle$ to obtain $y_j = \langle \phi | \hat{O}_j | \phi \rangle$
 - 2: Randomly initializing an MPS with bond dimension D
 - 3: Iteratively minimizing the loss function in Equation (10) using BFGS algorithm
-

3. Numerical Results

Now, we demonstrate our method by applying it to reconstruct several exemplary quantum states based on synthetic training data. We first consider the W state, which is a common state used to test the QST schemes. An n -qubit W state is defined as

$$|W_n\rangle = \frac{1}{\sqrt{n}}(|100\dots 0\rangle + |010\dots 0\rangle + \dots + |000\dots 1\rangle). \quad (13)$$

$|W_n\rangle$ can be written as an MPS with $D = n$. Taking the case $n = 3$ as an example, the coefficients of $|W_3\rangle$ can be compactly written as the following MPS:

$$\text{MPS}_W = [|1\rangle |0\rangle |0\rangle] \otimes \begin{bmatrix} |0\rangle & 0 & 0 \\ 0 & |1\rangle & 0 \\ 0 & 0 & |0\rangle \end{bmatrix} \otimes \begin{bmatrix} |0\rangle \\ |0\rangle \\ |1\rangle \end{bmatrix}, \quad (14)$$

where we can clearly see that the bond dimension of MPS_W is 3. The generalization of Equation (14) to general n is straightforward.

We first take $n = 4$ and evaluate the convergence of our method with the number iterations (epochs), which is shown in Figure 2. We have used 5 independently generated training datasets to avoid any bias of training data, where each dataset contains 50 samples (compared to the whole number of independent samples $4^4 = 256$). We have also used the Broyden–Fletcher–Goldfarb–Shanno (BFGS) optimizer for all of the simulations in this work. We can see that our method converges very quickly for all the different groups of training datasets; concretely, the loss values for all the simulations drop to around 10^{-9} within 20 iterations. However, the average fidelity (defined later in Equation (15)) of the reconstructed quantum states is only slightly higher than 60% since the number of samples is too small; nevertheless, this issue can be overcome by enlarging the number of samples, as will be shown later.

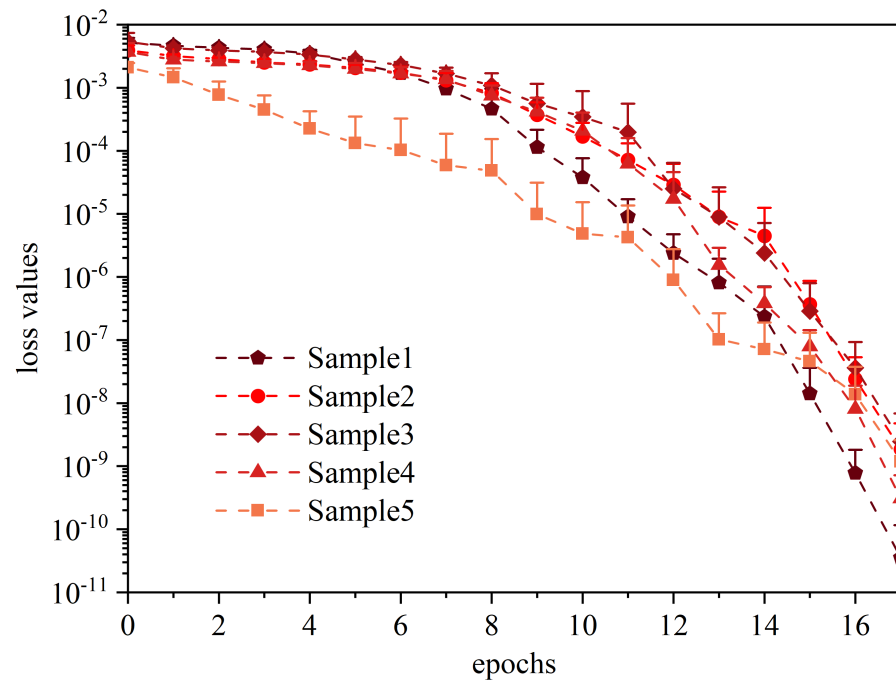


Figure 2. Convergence of the loss values towards the minimum with the number of iterations. The different lines correspond to five sets of independently generated training data with size $N_s = 50$. The error bar shows the standard derivation with respect to the mean value for 10 different runs, with random initialization for each training dataset.

Since we use much less data than the whole number of operators, it is possible that many different $|\tilde{\phi}\rangle$ s could give the same predictions y_j used in the training data, and in different runs we may obtain different states (since we use random initialization for $|\tilde{\phi}\rangle$ with a fixed bond dimension D in our simulations). To evaluate the overall quality of the reconstructed quantum state, we thus compute the average quantum fidelities by running the simulation for a given training dataset for 100 times, where the quantum fidelity $\mathcal{F}$ between two quantum states τ and σ is defined as [41]

$$\mathcal{F}(\tau, \sigma) = \left[\text{tr} \left(\sqrt{\sqrt{\tau} \sigma \sqrt{\tau}} \right) \right]^2. \quad (15)$$

The results of this simulation are shown in Figure 3, where we have considered both the W state and the GHZ state. We can see that when the number of samples N_s is not large enough, there is a significant chance that this will result in a wrong state compared to the target quantum state; however, as N_s increases, different runs could converge to the target state (the minimal N_s required to reach a certain fidelity for a given quantum state is studied later). Concretely, we can see that for the W state with $n = 4$ considered, $N_s = 80$ is enough to ensure most of the simulations using a different training dataset and with different initialization to converge to the correct state (except for one training dataset, whose average quantum fidelity is only around 60% for $N_s = 80$), while for the GHZ state, $N_s = 100$ is required to reach the same precision. In all these simulations, the loss values have converged to lower than 10^{-8} .

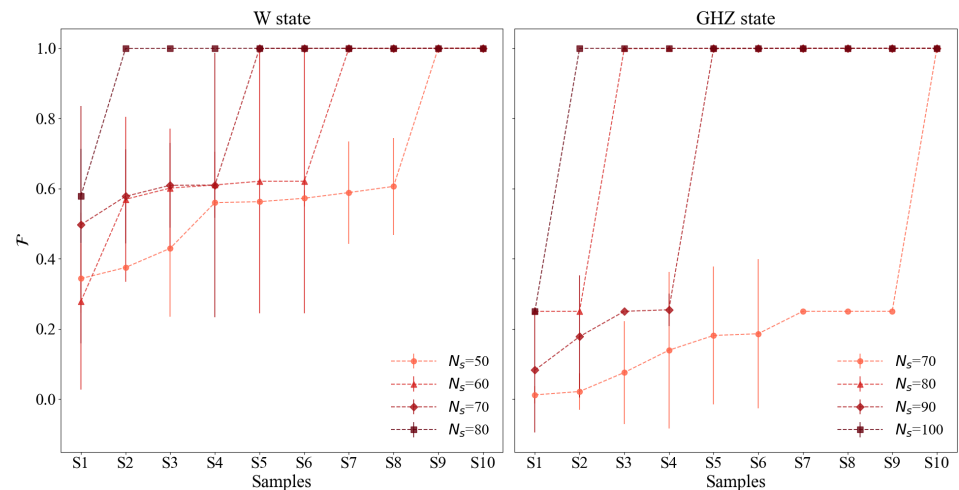


Figure 3. Average quantum fidelity between the reconstructed quantum state and the target quantum state for 100 random initialization. The x-axis corresponds to 10 independently generated train datasets. The dashed lines with a circle, triangle, diamond, and square represent the average values over the 100 random initialization corresponding to $N_s = 50, 60, 70$, and 80 , respectively. The error bars represent the standard derivations around the mean value.

Finally, we study the scaling of the minimal number of samples in the training dataset, denoted as N_s^* , such that 90% of the randomly generated training dataset can result in an average quantum fidelity above 95% (we still use 10 training datasets and 100 random initialization for each simulation), against the number of qubits n , which is shown in Figure 4. From Figure 4a, we can see that for W states N_s^* still scales exponentially with n ; in fact, N_s^* is approximately 1/4 of the total number of samples (the black solid line). In Figure 4b, we also study randomly generated MPSs, which may be generated using random quantum circuits in the experiment. We can see that in this case N_s^* scales sub-exponentially for the different bond dimensions considered. Interestingly, although W states seem to be much simpler compared to a randomly generated MPS, in our approach we can construct the latter class of states much better than the former.

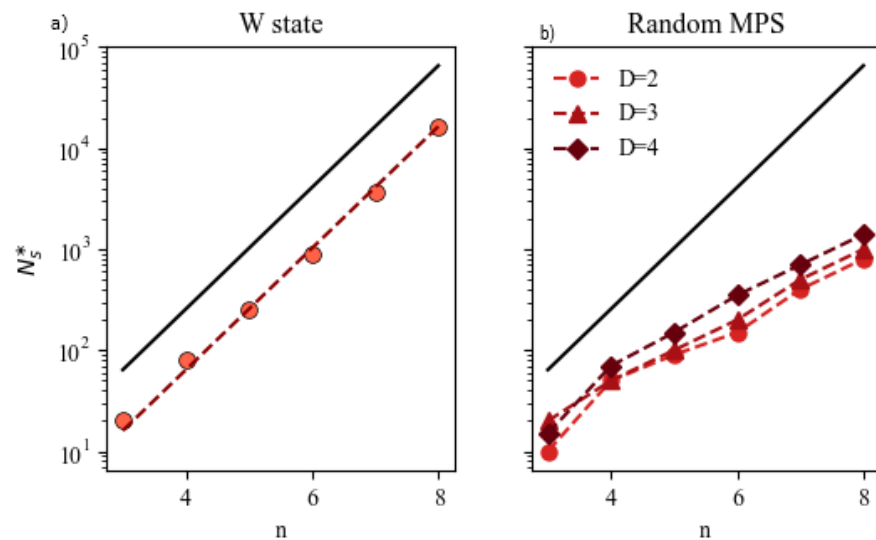


Figure 4. Minimum number of samples N_s^* required to accurately reconstruct the target quantum state as a function of n for (a) the W states and (b) the randomly generated matrix product states. The solid black line represents the size of the complete basis set. The dashed line in (a) is a linear fit of the average quantum fidelity for the W states. The dashed lines with circle, triangle, and diamond represent results for randomly generate MPSs with $D = 2, 3$, and 4 , respectively.

4. Conclusions

In this work, we have proposed a matrix-product-states-based supervised machine learning algorithm for the reconstruction of pure quantum states. The training data for our algorithm are the expectation values of a set of randomly chosen Pauli strings. Compared to the standard QST methods, our method could reconstruct the unknown quantum state using only a portion of measurements (expectation values) out of all of the possible ones. We have used this method to reconstruct the W states and randomly generated MPSs of different sizes. For W states, we show that only about 1/4 of the whole number of basis sets are required for accurate reconstruction, while randomly generated MPSs with fixed bond dimensions can be efficiently reconstructed with sub-exponential scaling against the number of qubits. Compared to existing scalable QST methods, our method is likely to require a larger number of training data; however, our method could always result in vanishing loss values (which means that the reconstructed quantum states will have exactly the same expectation values on the given set of measurement operators as the target quantum state) and is likely to have a high precision close to the standard QST when enough training datasets are given. The numerical experiments show that our method could be a suitable tool for high-precision QST with around 10 qubits.

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References

1. White, A.G.; James, D.F.V.; Eberhard, P.H.; Kwiat, P.G. Nonmaximally Entangled States: Production, Characterization, and Utilization. *Phys. Rev. Lett.* **1999**, *83*, 3103–3107. [\[CrossRef\]](#)
2. James, D.F.V.; Kwiat, P.G.; Munro, W.J.; White, A.G. Measurement of qubits. *Phys. Rev. A* **2001**, *64*, 052312. [\[CrossRef\]](#)
3. Gross, D.; Liu, Y.K.; Flammia, S.T.; Becker, S.; Eisert, J. Quantum State Tomography via Compressed Sensing. *Phys. Rev. Lett.* **2010**, *105*, 150401. [\[CrossRef\]](#) [\[PubMed\]](#)
4. Liu, W.T.; Zhang, T.; Liu, J.Y.; Chen, P.X.; Yuan, J.M. Experimental Quantum State Tomography via Compressed Sampling. *Phys. Rev. Lett.* **2012**, *108*, 170403. [\[CrossRef\]](#) [\[PubMed\]](#)
5. Smith, A.; Riofrío, C.A.; Anderson, B.E.; Sosa-Martinez, H.; Deutsch, I.H.; Jessen, P.S. Quantum state tomography by continuous measurement and compressed sensing. *Phys. Rev. A* **2013**, *87*, 030102. [\[CrossRef\]](#)
6. Riofrío, C.A.; Gross, D.; Flammia, S.T.; Monz, T.; Nigg, D.; Blatt, R.; Eisert, J. Experimental quantum compressed sensing for a seven-qubit system. *Nat. Commun.* **2017**, *8*, 15305. [\[CrossRef\]](#)
7. Tóth, G.; Wieczorek, W.; Gross, D.; Krischek, R.; Schwemmer, C.; Weinfurter, H. Permutationally Invariant Quantum Tomography. *Phys. Rev. Lett.* **2010**, *105*, 250403. [\[CrossRef\]](#)
8. Moroder, T.; Hyllus, P.; Tóth, G.; Schwemmer, C.; Niggebaum, A.; Gaile, S.; Gühne, O.; Weinfurter, H. Permutationally invariant state reconstruction. *New J. Phys.* **2012**, *14*, 105001. [\[CrossRef\]](#)
9. Liu, Y.; Wang, D.; Xue, S.; Huang, A.; Fu, X.; Qiang, X.; Xu, P.; Huang, H.L.; Deng, M.; Guo, C.; et al. Variational quantum circuits for quantum state tomography. *Phys. Rev. A* **2020**, *101*, 052316. [\[CrossRef\]](#)
10. Xue, S.; Liu, Y.; Wang, Y.; Zhu, P.; Guo, C.; Wu, J. Variational quantum process tomography of unitaries. *Phys. Rev. A* **2022**, *105*, 032427. [\[CrossRef\]](#)

11. Cramer, M.; Plenio, M.B.; Flammia, S.T.; Somma, R.; Gross, D.; Bartlett, S.D.; Landon-Cardinal, O.; Poulin, D.; Liu, Y.K. Efficient quantum state tomography. *Nat. Commun.* **2010**, *1*, 149. [\[CrossRef\]](#)
12. Baumgratz, T.; Gross, D.; Cramer, M.; Plenio, M.B. Scalable Reconstruction of Density Matrices. *Phys. Rev. Lett.* **2013**, *111*, 020401. [\[CrossRef\]](#)
13. Lanyon, B.P.; Maier, C.; Holzäpfel, M.; Baumgratz, T.; Hempel, C.; Jurcevic, P.; Dhand, I.; Buyskikh, A.S.; Daley, A.J.; Cramer, M.; et al. Efficient tomography of a quantum many-body system. *Nat. Phys.* **2017**, *13*, 1158–1162. [\[CrossRef\]](#)
14. Guo, C. Reconstructing non-Markovian open quantum evolution from multiple time measurements. *Phys. Rev. A* **2022**, *106*, 022411. [\[CrossRef\]](#)
15. Zhao, Y.Y.; Hou, Z.; Xiang, G.Y.; Han, Y.J.; Li, C.F.; Guo, G.C. Experimental demonstration of efficient quantum state tomography of matrix product states. *Opt. Express* **2017**, *25*, 9010–9018. [\[CrossRef\]](#)
16. Wang, J.; Han, Z.Y.; Wang, S.B.; Li, Z.; Mu, L.Z.; Fan, H.; Wang, L. Scalable quantum tomography with fidelity estimation. *Phys. Rev. A* **2020**, *101*, 032321. [\[CrossRef\]](#)
17. Kurmapu, M.K.; Tiunova, V.; Tiunov, E.; Ringbauer, M.; Maier, C.; Blatt, R.; Monz, T.; Fedorov, A.K.; Lvovsky, A. Reconstructing complex states of a 20-qubit quantum simulator. *arXiv* **2022**, arXiv:2208.04862.
18. Han, D.; Guo, C.; Wang, X. Density matrix reconstruction using non-negative matrix product states. *Phys. Rev. A* **2022**, *106*, 042435. [\[CrossRef\]](#)
19. Lidiak, A.; Jameson, C.; Qin, Z.; Tang, G.; Wakin, M.B.; Zhu, Z.; Gong, Z. Quantum state tomography with tensor train cross approximation. *arXiv* **2022**, arXiv:2207.06397.
20. Torlai, G.; Mazzola, G.; Carrasquilla, J.; Troyer, M.; Melko, R.; Carleo, G. Neural-network quantum state tomography. *Nat. Phys.* **2018**, *14*, 447–450. [\[CrossRef\]](#)
21. Torlai, G.; Melko, R.G. Latent Space Purification via Neural Density Operators. *Phys. Rev. Lett.* **2018**, *120*, 240503. [\[CrossRef\]](#) [\[PubMed\]](#)
22. Xin, T.; Lu, S.; Cao, N.; Anikeeva, G.; Lu, D.; Li, J.; Long, G.; Zeng, B. Local-measurement-based quantum state tomography via neural networks. *NPJ Quantum Inf.* **2019**, *5*, 109. [\[CrossRef\]](#)
23. Palmieri, A.M.; Kovlakov, E.; Bianchi, F.; Yudin, D.; Straupe, S.; Biamonte, J.D.; Kulik, S. Experimental neural network enhanced quantum tomography. *NPJ Quantum Inf.* **2020**, *6*, 20. [\[CrossRef\]](#)
24. Gray, J.; Banchi, L.; Bayat, A.; Bose, S. Machine-Learning-Assisted Many-Body Entanglement Measurement. *Phys. Rev. Lett.* **2018**, *121*, 150503. [\[CrossRef\]](#) [\[PubMed\]](#)
25. Rocchetto, A.; Grant, E.; Strelchuk, S.; Carleo, G.; Severini, S. Learning hard quantum distributions with variational autoencoders. *NPJ Quantum Inf.* **2018**, *4*, 28. [\[CrossRef\]](#)
26. Yu, S.; Albarrán-Arriagada, F.; Retamal, J.C.; Wang, Y.T.; Liu, W.; Ke, Z.J.; Meng, Y.; Li, Z.P.; Tang, J.S.; Solano, E.; et al. Reconstruction of a Photonic Qubit State with Reinforcement Learning. [\[CrossRef\]](#)
27. Melkani, A.; Gneiting, C.; Nori, F. Eigenstate extraction with neural-network tomography. *Phys. Rev. A* **2020**, *102*, 022412. [\[CrossRef\]](#)
28. Weiss, T.; Romero-Isart, O. Quantum motional state tomography with nonquadratic potentials and neural networks. *Phys. Rev. Res.* **2019**, *1*, 033157. [\[CrossRef\]](#)
29. Xin, T.; Nie, X.; Kong, X.; Wen, J.; Lu, D.; Li, J. Quantum Pure State Tomography via Variational Hybrid Quantum-Classical Method. *Phys. Rev. Appl.* **2020**, *13*, 024013. [\[CrossRef\]](#)
30. Carrasquilla, J.; Torlai, G.; Melko, R.G.; Aolita, L. Reconstructing quantum states with generative models. *Nat. Mach. Intell.* **2019**, *1*, 155–161. [\[CrossRef\]](#)
31. Ahmed, S.; Sánchez Muñoz, C.; Nori, F.; Kockum, A.F. Quantum State Tomography with Conditional Generative Adversarial Networks. *Phys. Rev. Lett.* **2021**, *127*, 140502. [\[CrossRef\]](#)
32. Aaronson, S. Shadow Tomography of Quantum States. In Proceedings of the 50th Annual ACM SIGACT Symposium on Theory of Computing, STOC 2018, Los Angeles, CA, USA, 25–29 June 2018; Association for Computing Machinery: New York, NY, USA, 2018; pp. 325–338. [\[CrossRef\]](#)
33. Huang, H.Y.; Kueng, R.; Preskill, J. Predicting many properties of a quantum system from very few measurements. *Nat. Phys.* **2020**, *16*, 1050–1057. [\[CrossRef\]](#)
34. Chen, S.; Yu, W.; Zeng, P.; Flammia, S.T. Robust Shadow Estimation. *PRX Quantum* **2021**, *2*, 030348. [\[CrossRef\]](#)
35. Gupta, R.; Xia, R.; Levine, R.D.; Kais, S. Maximal Entropy Approach for Quantum State Tomography. *PRX Quantum* **2021**, *2*, 010318. [\[CrossRef\]](#)
36. Endo, S.; Benjamin, S.C.; Li, Y. Practical Quantum Error Mitigation for Near-Future Applications. *Phys. Rev. X* **2018**, *8*, 031027. [\[CrossRef\]](#)
37. Endo, S.; Cai, Z.; Benjamin, S.C.; Yuan, X. Hybrid Quantum-Classical Algorithms and Quantum Error Mitigation. *J. Phys. Soc. Jpn.* **2021**, *90*, 032001. [\[CrossRef\]](#)
38. Peng, Z.; De Graaf, S.; Tsai, J.; Astafiev, O. Tuneable on-demand single-photon source in the microwave range. *Nat. Commun.* **2016**, *7*, 1–6. [10.1038/ncomms12588](#). [\[CrossRef\]](#)
39. Schollwöck, U. The density-matrix renormalization group in the age of matrix product states. *Ann. Phys.* **2011**, *326*, 96–192. [\[CrossRef\]](#)

40. Guo, C.; Poletti, D. Scheme for automatic differentiation of complex loss functions with applications in quantum physics. *Phys. Rev. E* **2021**, *103*, 013309. [[CrossRef](#)]
41. Nielsen, M.A.; Chuang, I. Quantum Computation and Quantum Information. *Am. J. Phys.* **2002**, *70*, 558. [[CrossRef](#)]

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